INTERIM AMENDMENT
FEDERAL SPECIFICATION

HAIR, ANIMAL, CURLED

This interim amendment was developed by the U. S. Army Natick Laboratories (GL) Natick, Massachusetts 01760, for use by Defense Personnel Support Center, based on currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim amendment as a valid exception to Federal Specification C-H-111d, dated February 15, 1968.

Page 2, Add new paragraph:

"3.2.1 The hair, after being thoroughly washed, shall be treated so as to contain no less than 0.50 percent nor more than 0.70 percent DDT when treated with a DDT solvent solution or water emulsion. The DDT content shall be based on the air dried weight of the treated hair when tested as specified in 4.3.4.3."

Page 5, Add new paragraph:

"4.3.4.3 DDT (dichloro-diphenyl-trichloroethane) content. A 10-gram plus or minus 1-gram quantity of natural hair shall be extracted for two hours in a Soxhlet apparatus with petroleum ether or any other suitable solvent, after which most of the solvent shall be removed by evaporation on a steam bath. Do not evaporate to dryness, since the DDT may decompose. Add 25 ml. of 99 percent isopropyl alcohol and 2.5 grams of metallic sodium cut into small pieces and swirl the flask in order to mix its contents. Connect to a water-cooled reflux condenser and boil gently for a minimum of one half hour. Shake the flask occasionally. Decompose the excess sodium by cautiously adding 10 ml. of 50 percent isopropyl alcohol through the condenser at a rate of one to two drops per second. Boil for an additional 10 minutes and then add 60 ml. of distilled water. Cool to room temperature, add two to three drops of phenolphthalein solution, neutralize by adding nitric acid (1 to 1), and add 10 ml. of the diluted acid in excess. Add dropwise, with stirring of the solution, a measured excess (25 ml.) of 0.1N $AgNO_3$ solution. Coagulate the precipitate by heating on a steam bath for approximately one half hour. Cool to room temperature and filter through a No. 52 Whatman filter paper and wash thoroughly with distilled water, receiving the filtrate in a 500-ml. Erlenmeyer flask. Add 5 ml. of ferric ammonium sulfate indicator and titrate the excess $AgNO_3$ with 0.1N KSCN. Compute the net number of ml. of 0.1N $AgNO_3$ consumed by the sample. Calculate percent DDT as follows:

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Percent DDT (dichloro-diphenyl-tricholorethane) =

$$\frac{\text{Milliliters of 0.1N AgNo (consumed) x .7094}}{\text{Weight of sample (grams)}}$$

A blank determination for the chemicals used shall be made following the exact procedure given above, but limiting the 0.1N $AgNO_3$ solution to 5 ml. in order to obtain a chloride correction value. This blank value shall be subtracted from the DDT determination to obtain the exact value of DDT present."

MILITARY CUSTODIAN:

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Preparing activity:

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Project No. 8320-A051

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